Theory of Elementary Particles
Lecture 14

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Majorana Particle

Majorana Neutrino

See-Saw Mechanism

Neutrinoless double-beta decay

Extending Standard Model by Neutrino Mass Term

Neutrino oscillations

Majorana Particle

- A mass term of the so-called Dirac Particle:

$$-m\overline{\psi}\psi = -m(\overline{\psi_R}\psi_L + \overline{\psi_L}\psi_R) \tag{1}$$

- If only the left-handed neutrino exists, then the Dirac mass term vanishes and the neutrino must be massless

- Let us consider Lorentz invariant Lagrangian describing neutral particles
- Bi-spinor transforms as

$$\psi'(x') = S\psi(x) \quad S = \exp\left(-\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right), \quad x'^{\mu} = \Lambda^{\mu}{}_{\nu}x^{\nu} \tag{2}$$

where

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^{\mu}, \gamma^{\nu}] \tag{3}$$

Notice that

$$\sigma^{\mu\nu\dagger} = -\frac{i}{2}[\gamma^{\dagger}_{\nu}, \gamma^{\dagger}_{\mu}] = \frac{i}{2}[\gamma^{\dagger}_{\mu}, \gamma^{\dagger}_{\nu}] \tag{4}$$

we see that

$$\sigma^{kn\dagger} = \sigma^{kn}, \quad \sigma^{0n\dagger} = -\sigma^{0n}, \quad \sigma^{\mu\nu\dagger} = \gamma_0\sigma^{\mu\nu}\gamma_0 \tag{5}$$

- Now, notice that

$$\overline{\psi'} = \psi'^{\dagger}\gamma_0 = \psi^{\dagger}\exp\left(\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma^{\dagger}_{\mu\nu}\right)\gamma_0 \tag{6}$$

$$= \psi^{\dagger}\gamma_0\exp\left(\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\gamma_0\sigma^{\dagger}_{\mu\nu}\gamma_0\right) \tag{7}$$

$$= \psi^{\dagger}\gamma_0\exp\left(\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right) = \overline{\psi}\exp\left(\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right) = \overline{\psi}S^{-1} \tag{8}$$

- Hence

$$\overline{\psi'}\psi' = \overline{\psi}\exp\left(\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right)\exp\left(-\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right)\psi = \overline{\psi}\psi \tag{9}$$

- Now $\gamma_5\sigma_{\mu\nu} = \sigma_{\mu\nu}\gamma_5$

$$\psi_L \to \frac{1}{2}(1-\gamma_5)\exp\left(-\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right)\psi = \exp\left(-\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right)\psi_L \tag{10}$$

Similarly

$$\psi_R \to \frac{1}{2}(1+\gamma_5)\exp\left(-\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right)\psi = \exp\left(-\frac{i}{4}\epsilon^{\mu\nu}(\Lambda)\sigma_{\mu\nu}\right)\psi_R \tag{11}$$

* Left-handed and Right-handed components transform independently!

- Consider a field operator which is charged conjugated

$$\psi^C = \mathcal{C}\psi\mathcal{C}^{-1} = i\gamma_2\gamma^0\overline{\psi}^T = i\gamma_2\psi^* = C\,\overline{\psi}^T, \quad C = i\gamma_2\gamma^0 \tag{12}$$

- Properties

$$C^\dagger = i\gamma^0\gamma_2, \quad C^\dagger C = i\gamma^0\gamma_2 i\gamma_2\gamma^0 = (-)(-) = 1 \tag{13}$$

- Notice that

$$C^\dagger\gamma^\mu C = -\gamma^{\mu T} \tag{14}$$

indeed for $\mu = 0$ and 2 $C^\dagger\gamma^\mu C = -\gamma^\mu$ and for $\mu = 1$ and 3 $C^\dagger\gamma^\mu C = +\gamma^\mu$

- Properties

$$C^T = i\gamma_0\gamma_2 = C^\dagger \tag{15}$$

and

$$C^{-1} = C^\dagger = C^T = -C \tag{16}$$

- Then

$$
\begin{aligned}
(S\psi)^C &= C\left(\overline{S\psi}\right)^T && (17)\\
&= C\left(\overline{\psi}\,S^{-1}\right)^T = C\left(S^{-1}\right)^T\overline{\psi}^T && (18)\\
&= C\left(S^{-1}\right)^T C^{-1}\,C\overline{\psi}^T && (19)\\
&= \left(C\left(S^{-1}\right)^T C^{-1}\right)\psi^C && (20)\\
&= S\psi^C && (21)
\end{aligned}
$$

where we used the property

$$
C\,\sigma^T_{\mu\nu}\,C^{-1} = -\sigma_{\mu\nu} \quad \left(\text{from } C^{-1}\gamma^\mu C = -\gamma^{\mu T}\right), \quad \text{hence} \quad C\left(S^{-1}\right)^T C^{-1} = S \tag{22}
$$

- We see that field ψ^C transforms the same as ψ!
- Then we may have four combinations for forming mass terms

$$
((..)\overline{\psi} + (..)\overline{\psi^C})((..)\psi + (..)\psi^C) \tag{23}
$$

- Indeed contributions:

$$\overline{\psi^C}\psi^C, \quad \overline{\psi^C}\psi, \quad \overline{\psi}\psi^C \tag{24}$$

  are also Lorentz invariant!

- However, (see later)

$$\overline{\psi^C}\psi^C \to \overline{\psi}\psi$$

  so only red contributions matter

- Notice that

$$\begin{aligned} \psi_L^C &= \mathcal{C}(\psi_L)\mathcal{C}^{-1} = i\gamma_2\psi_L^* \qquad (25)\\ &= i\gamma_2\frac{1}{2}(1-\gamma_5^*)\psi^* = i\gamma_2\frac{1}{2}(1-\gamma_5)\psi^* = \frac{1}{2}(1+\gamma_5)i\gamma_2\psi^* \\ &= \frac{1}{2}(1+\gamma_5)\psi^C = (\psi^C)_R \qquad (26)\end{aligned}$$

- Analogously

$$\psi_R^C = (\psi^C)_L \qquad (27)$$

- then

$$\begin{aligned} \overline{\psi_L^C} &= \overline{\frac{1}{2}(1+\gamma_5)\psi^C} = (\psi^C)^\dagger\frac{1}{2}(1+\gamma_5)\gamma_0 = (\psi^C)^\dagger\gamma_0\frac{1}{2}(1-\gamma_5) = \overline{\psi^C}\frac{1}{2}(1-\gamma_5) \\ &= (\overline{\psi^C})_L \qquad (28)\end{aligned}$$

- Analogously

$$\overline{\psi_R^C} = (\overline{\psi^C})_R \qquad (29)$$

$$\underbrace{\psi_L, \quad \psi_R}, \quad \underbrace{(\psi^C)_L, \quad (\psi^C)_R} \tag{30}$$

as well as

$$\underbrace{\overline{\psi_L}, \quad \overline{\psi_R}}, \quad \underbrace{\overline{(\psi^C)_L}, \quad \overline{(\psi^C)_R}} \tag{31}$$

$$
\begin{aligned}
\overline{(\psi^C)_L}(\psi^C)_R &= \overline{(\psi_R)^C}(\psi_L)^C = (\mathrm{i}\gamma_2\psi_R^*)^\dagger\gamma_0\mathrm{i}\gamma_2\psi_L^* \\
&= -\psi_R^T\gamma_2\gamma_0\gamma_2\psi_L^* = -\psi_R^T\gamma_0\psi_L^* \\
&= \overline{\psi_L}\psi_R \quad \text{(anticommuting fields)} \qquad (32)
\end{aligned}
$$

Then we have possible combinations:

$$\overline{(\psi^C)_L}\psi_R \leftarrow h.c. \rightarrow \overline{\psi_R}(\psi^C)_L, \tag{33}$$

$$\overline{(\psi^C)_R}\psi_L \leftarrow h.c. \rightarrow \overline{\psi_L}(\psi^C)_R \tag{34}$$

where we used the property that

$$(\overline{(\psi^C)_L}\psi_R)^\dagger = \overline{\psi_R}(\psi^C)_L. \tag{35}$$

Finally, the mass terms are given by combinations of three bilinear forms

$$\overline{\psi_L}\psi_R, \quad \overline{(\psi^C)_L}\psi_R, \quad \overline{(\psi^C)_R}\psi_L \quad + h.c. \tag{36}$$

The most general Lorentz-invariant mass term

$$\mathcal{L}_m = m_D \left(\overline{\psi_L}\psi_R + h.c.\right) + m_L \left(\overline{(\psi^C)_R}\psi_L + h.c.\right) + m_R \left(\overline{(\psi^C)_L}\psi_R + h.c.\right) \quad (37)$$

Notice that the last two terms violate global $U(1)$ symmetry, hence lepton number conservation!It can be written in the form

$$\mathcal{L}_m = m_D \left(\overline{\psi_L}\psi_R + h.c.\right) + m_L \left(\overline{(\psi_L)^C}\psi_L + h.c.\right) + m_R \left(\overline{(\psi_R)^C}\psi_R + h.c.\right). \quad (38)$$

Majorana Neutrino

$$\nu^C = \nu \tag{39}$$

It works only for neutral particles (a Majorana mass violates the $U(1)$ charge symmetry)

Let us introduce the fields

$$\chi = \psi_L + (\psi_L)^C, \quad \chi^C = \chi \tag{40}$$
$$\omega = \psi_R + (\psi_R)^C, \quad \omega^C = \omega. \tag{41}$$

or inverse formulas

$$\psi_L = \frac{1}{2}(1-\gamma_5)\chi; \quad (\psi_L)^C = \frac{1}{2}(1+\gamma_5)\chi \tag{42}$$
$$\psi_R = \frac{1}{2}(1+\gamma_5)\omega; \quad (\psi_R)^C = \frac{1}{2}(1-\gamma_5)\omega \tag{43}$$

Notice that

$$\gamma_5 \begin{pmatrix} \psi \\ \chi \\ \omega \end{pmatrix} = \begin{pmatrix} \psi' \\ \chi' \\ \omega' \end{pmatrix} = \begin{pmatrix} -\psi_L + \psi_R \\ -\psi_L + (\psi_L)^C \\ +\psi_R - (\psi_R)^C \end{pmatrix} \tag{44}$$

It redefines the properties of the fields.

After redefinitions of fields, the mass Lagrangian reads

$$\begin{aligned}\mathcal{L}_m &= \frac{m_D}{2}(\overline{\chi}\omega + \overline{\omega}\chi) + m_L\overline{\chi}\chi + m_R\overline{\omega}\omega \\ &= \frac{1}{2}\begin{pmatrix}\overline{\chi} & \overline{\omega}\end{pmatrix}\begin{pmatrix}2m_L & m_D \\ m_D & 2m_R\end{pmatrix}\begin{pmatrix}\chi \\ \omega\end{pmatrix}. \end{aligned} \tag{45}$$

Diagonalization leads to two mass eigenvalues

$$M_\pm = m_L + m_R \pm \sqrt{(m_L - m_R)^2 + m_D^2}. \tag{46}$$

The corresponding eigenfields are

$$\eta_1 = \chi\cos\theta - \omega\sin\theta, \tag{47}$$

$$\eta_2 = \chi\sin\theta + \omega\cos\theta, \tag{48}$$

where

$$\tan(2\theta) = \frac{m_D}{m_R - m_L}. \tag{49}$$

The physical masses are $|M_\pm|$.

- Consider the case in which Majorana masses disappear, $m_{L,R} = 0$, but the Dirac mass term $m_D \neq 0$. It corresponds to $\theta = \pi/4$.
- Then there are two eigenstates

$$\frac{1}{\sqrt{2}}(\chi \pm \omega) \tag{50}$$

with eigenvalues $\pm m_D$.

- One eigenvalue is negative. A chiral transformation of the corresponding Majorana field changes the sign of its mass term.

$$\frac{1}{\sqrt{2}}(\chi + \omega) = \frac{1}{\sqrt{2}}\left(\psi_L + (\psi_L)^C + \psi_R + (\psi_R)^C\right) \equiv \xi_1. \tag{51}$$

The second state after transformation:

$$\frac{1}{\sqrt{2}}(\chi' - \omega') = \frac{1}{\sqrt{2}}\left(-\psi_L + (\psi_L)^C - \psi_R + (\psi_R)^C\right) \equiv \xi_2. \tag{52}$$

- The mass term is invariant under orthogonal rotations in the (ξ_1, ξ_2) plane:

$$\mathcal{L}_m = \frac{m_D}{2}\left(\overline{\xi}_1\xi_1 + \overline{\xi}_2\xi_2\right) = \frac{m_D}{2}\left(\overline{\xi}'_1\xi'_1 + \overline{\xi}'_2\xi'_2\right). \tag{53}$$

For $\theta = \pi/4$ we obtain

$$\xi_1' = \frac{1}{\sqrt{2}}(\xi_1 - \xi_2) = \psi_L + \psi_R = \psi, \tag{54}$$

$$\xi_2' = \frac{1}{\sqrt{2}}(\xi_1 + \xi_2) = (\psi_L)^C + (\psi_R)^C = \psi^C. \tag{55}$$

Hence, the mass term becomes

$$\mathcal{L}_m = m_D \overline{\psi}\psi. \tag{56}$$

Remark

A Dirac particle (the limit $m_{L,R} = 0$) can be described as a combination of two degenerate Majorana particles.

See-Saw Mechanism

- Let us consider $m_L = 0$, as well as

$$m_D \ll m_R. \tag{57}$$

- The mass eigenvalues read

$$\begin{aligned} M_{1,2} &= m_R \left(1 \pm \sqrt{1 + \frac{m_D^2}{m_R^2}} \right) \\ &\approx m_R \left(1 \pm \left(1 + \frac{m_D^2}{2m_R^2} \right) \right). \end{aligned} \tag{58}$$

- Then

$$M_1 \approx 2m_R, \tag{59}$$

$$M_2 \approx -\frac{m_D^2}{2m_R}. \tag{60}$$

- The physical masses are

$$m_{\text{heavy}} \approx 2m_R, \qquad m_{\text{light}} \approx \frac{m_D^2}{2m_R}. \tag{61}$$

- Let us consider two neutrinos: one predominantly right-handed (heavy) and the observed predominantly left-handed (light) neutrino.
- We assume that the Dirac mass parameter is of the order of the charged-lepton mass:

$$m_D \sim m_l. \tag{62}$$

- Using the see-saw approximation, we obtain the mass hierarchy

$$m_\nu^{\text{light}} \sim \frac{m_l^2}{2m_R} \ll m_l \ll m_\nu^{\text{heavy}} \simeq 2m_R. \tag{63}$$

Property

The small neutrino mass is related to the charged-lepton mass scale and the large mass of a predominantly right-handed neutrino, which interacts only very weakly with ordinary matter.

This mechanism is called the see-saw mechanism.

Neutrinoless double-beta decay

Let us consider the process, which may occur only for Majorana neutrinos,

$$(A, Z) \rightarrow (A, Z + 2) + 2e^{-} \tag{64}$$

Lepton number is violated by $\Delta L = 2$, and it is forbidden for Dirac neutrinos.
Matrix element reads:

$$S_{fi} = T \exp(\mathrm{i} \int d^4 x \mathcal{L}_{WSG}) \tag{65}$$

The tree level amplitude reads

$$\mathcal{A}_{\beta\beta 0\nu} = \frac{g^2}{2}\sum_m T\int d^4x d^4y$$
$$\left\langle e(k')e(p')\right| W^\mu(x)\overline{e}(x)\gamma_\mu(1-\gamma_5)U_{em}\nu_m(x)W^\nu(y)\overline{e}(y)\gamma_\nu(1-\gamma_5)U_{em}\nu_m(y)\left|W(k)W(p)\right\rangle \tag{66}$$

- Effectively we must consider contraction of two neutrino fields.
- If the neutrino is a Dirac particle then the contraction vanishes.
- For a Majorana particle the amplitude does not vanish.

Consider the current:

$$\begin{aligned}
\overline{e}\gamma_\mu(1-\gamma_5)\nu_m &= \overline{(e^c)^c}\gamma_\mu(1-\gamma_5)(\nu_m^c)^c && (67)\\
&= \overline{C\gamma_0 e^{c*}}\gamma_\mu(1-\gamma_5)C\gamma_0\nu_m^{c*} && (68)\\
&= e^{cT}\gamma_0 C^\dagger\gamma_0\gamma_\mu(1-\gamma_5)C\gamma_0\nu_m^{c*} && (69)
\end{aligned}$$

We use the property:

$$\begin{aligned}
C^\dagger &= C^{-1} && (70)\\
C^{-1}\gamma_\mu C &= -\gamma_\mu^T && (71)\\
\nu_m^c &= \lambda_m\nu_m && (72)
\end{aligned}$$

Then

$$\begin{aligned}
\overline{e}\gamma_\mu(1-\gamma_5)\nu_m &= \lambda_m e^{cT}\gamma_0 C^{-1}\gamma_0 C C^{-1}\gamma_\mu C C^{-1}(1-\gamma_5)C\gamma_0\nu_m^*\\
&= \lambda_m\, e^{cT}\gamma_\mu^T(1-\gamma_5)^T\gamma_0(\nu_m^\dagger)^T && (73)\\
&= \lambda_m\nu_m^\dagger\gamma_0(1-\gamma_5)\gamma_\mu e^c && (74)\\
&= \lambda_m\overline{\nu_m}\gamma_\mu(1+\gamma_5)e^c && (75)
\end{aligned}$$

Then we make contraction of currents with the Majorana propagator, which in momentum space reads:

$$\mathrm{i}\frac{\not{p}+M_m}{p^2-M_m^2} \tag{76}$$

Then the scattering amplitude in the momentum space reads

$$\mathcal{A}_{\beta\beta 0\nu} \quad = \quad \sum_m \left(W^\mu \overline{e}\gamma_\mu(1-\gamma_5)U_{em}\nu_m\right)\left(W^\nu \lambda_m \overline{\nu_m}\gamma_\nu(1+\gamma_5)e^c U_{em}\right) \tag{77}$$

Remembering what structure has electron field and assuming that W bosons are off shell external particles we replace the W_μ fields by their polarizations ϵ_i^μ, then

$$\mathcal{A}_{\beta\beta 0\nu} \quad = \quad \sum_m \epsilon_1^\mu \overline{u}_1\gamma_\mu(1-\gamma_5)\mathrm{i}\lambda_m \frac{\not{p}+M_m}{q^2-M_m^2}\not{\epsilon}_2(1+\gamma_5)\mathrm{i}\gamma_2(v(p,s))^* U_{em}^2 \tag{78}$$

Commuting $\not{p}$ through the chiral projectors (only the mass term survives) we get:

$$\mathcal{A}_{\beta\beta 0\nu} = 2\sum_m \lambda_m \overline{u}_1 \not{\epsilon}_1 \frac{M_m}{q^2}\not{\epsilon}_2(1-\gamma_5)\mathrm{i}\gamma_2(v(p,s))^* U_{em}^2 \tag{79}$$

Hence, schematically, the amplitude is proportional to:

$$\mathcal{A}_{\beta\beta 0\nu} \propto \sum_m \lambda_m M_m U_{em}^2 \tag{80}$$

Extending Standard Model by Neutrino Mass Term

- $$l_L = \begin{pmatrix} \nu \\ l^- \end{pmatrix} \sim (\underbrace{\mathbf{2}}_{SU(2) rep.dim.}, \underbrace{-1}_{Y}) \tag{81}$$

- $$\Phi = \begin{pmatrix} \Phi^+ \\ \Phi^0 \end{pmatrix} \sim (\mathbf{2}, +1) \tag{82}$$

- Possible mass term:

$$\begin{aligned} \overline{l_L} e_R &\sim (\mathbf{2}, 1) \times (\mathbf{1}, -2) = (\mathbf{2}, -1) && (83) \\ \overline{l_L^C} l_L &\sim (\mathbf{2}, -1) \times (\mathbf{2}, -1) = (\mathbf{1}, -2) + (\mathbf{3}, -2) && (84) \\ \overline{e_R^C} e_R &\sim (\mathbf{1}, -2) \times (\mathbf{1}, -2) = (\mathbf{1}, -4) && (85) \end{aligned}$$

- If $\phi = (\mathbf{2}, 1)$ only $\overline{l_L}\, \Phi\, e_R + h.c.$ is present in the Standard Model and lepton number is conserved!

Three possibilities to extend the Standard Model:

(1) Extension of the Higgs sector only

(2) Extension of the lepton sector only

(3) Extension of both Higgs and lepton sectors

Ad 1

One may introduce additional Higgs fields

- Triplet: $\mathbf{H} \sim (\mathbf{3}, 2)$

$$\boldsymbol{\tau} \cdot \mathbf{H} = \begin{pmatrix} H^+ & \sqrt{2}H^{++} \\ \sqrt{2}H^0 & -H^+ \end{pmatrix} \tag{86}$$

- Singlet: $h^+ \sim (\mathbf{1}, 2)$
- Double charged singlet: $R^{++} \sim (\mathbf{1}, 4)$

* $Q = \frac{Y}{2} + T_3$

Ad 2

- One introduces a neutral singlet, the right-handed (sterile) neutrino ν_R; then we have additional terms in the Lagrangian

$$\mathcal{L}' = f\overline{l_{iL}}\nu_R\Phi_i^* + B\overline{(\nu_R)^C}\nu_R + h.c. \rightarrow D\overline{\nu_L}\nu_R + B\overline{(\nu_R)^C}\nu_R \qquad (87)$$

Neutrino Oscillations

- Consider two-flavour case

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = U(\theta) \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix} \tag{88}$$

  where $\nu_{1,2}$ corresponds to $m_{1,2}$ mass eigenstate

- State of mass m_1 evolves in time as it follows

$$\mid \nu_1(t)\rangle = e^{-iE_1 t} \mid \nu_1(0)\rangle \tag{89}$$

  where $E_j^2 = \mathbf{p}_j^2 + m_j^2$ for relativistic neutrinos

- For relativistic neutrinos $m_j \ll |\mathbf{p}_j|$ and assuming that neutrinos of different flavor have the same momentum $|\mathbf{p}_1| = |\mathbf{p}_2| = E$ and

$$E_j \approx E + \frac{m_j^2}{2E} \tag{90}$$

- On the other hand

$$\begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} \tag{91}$$

Then

$$\nu_e(0) = \cos\theta\nu_1(0) + \sin\theta\nu_2(0) \quad (92)$$
$$\nu_\mu(0) = -\sin\theta\nu_1(0) + \cos\theta\nu_2(0) \quad (93)$$

and

$$\nu_1(0) = \cos\theta\nu_e(0) - \sin\theta\nu_\mu(0) \quad (94)$$
$$\nu_2(0) = \sin\theta\nu_e(0) + \cos\theta\nu_\mu(0) \quad (95)$$

$$|\,\nu_e(t)\rangle = \cos\theta e^{-iE_1 t}\,|\,\nu_1(0)\rangle + \sin\theta e^{-iE_2 t}\,|\,\nu_2(0)\rangle \quad (96)$$
$$|\,\nu_\mu(t)\rangle = -\sin\theta e^{-iE_1 t}\,|\,\nu_1(0)\rangle + \cos\theta e^{-iE_2 t}\,|\,\nu_2(0)\rangle \quad (97)$$

and finally

$$\begin{aligned}|\,\nu_\mu(t)\rangle &= -\sin\theta e^{-iE_1 t}\left(\cos\theta\,|\,\nu_e(0)\rangle - \sin\theta\,|\,\nu_\mu(0)\rangle\right)\\ &\quad + \cos\theta e^{-iE_2 t}\left(\sin\theta\,|\,\nu_e(0)\rangle + \cos\theta\,|\,\nu_\mu(0)\rangle\right) \quad (98)\\ &= \sin\theta\cos\theta(e^{-iE_2 t} - e^{-iE_1 t})\,|\,\nu_e(0)\rangle\\ &\quad + \left(\sin^2\theta e^{-iE_1 t} + \cos^2\theta e^{-iE_2 t}\right)\,|\,\nu_\mu(0)\rangle \quad (99)\end{aligned}$$

$$
\begin{aligned}
\mid \nu_\mu(t)\rangle &= (e^{-iE_1 t}\sin^2\theta + e^{-iE_2 t}\cos^2\theta) \mid \nu_\mu(0)\rangle + \cos\theta\sin\theta(e^{-iE_2 t} - e^{-iE_1 t}) \mid \nu_e(0)\rangle \\
\mid \nu_e(t)\rangle &= (e^{-iE_1 t}\cos^2\theta + e^{-iE_2 t}\sin^2\theta) \mid \nu_e(0)\rangle + \cos\theta\sin\theta(e^{-iE_2 t} - e^{-iE_1 t}) \mid \nu_\mu(0)\rangle
\end{aligned}
$$

easily we get

$$
\begin{aligned}
\langle \nu_\mu(0) \mid \nu_\mu(t)\rangle &= \left(\sin^2\theta e^{-iE_1 t} + \cos^2\theta e^{-iE_2 t}\right) && (100)\\
&= e^{-iE_1 t}\left(\sin^2\theta + \cos^2\theta e^{-i(E_2-E_1)t}\right) && (101)\\
\mid \langle \nu_\mu(0) \mid \nu_\mu(t)\rangle \mid^2 &= \sin^4\theta + \cos^4\theta + 2\sin^2\theta\cos^2\theta\cos(E_2 - E_1)t && (102)\\
&= \sin^4\theta + \cos^4\theta && \\
&\quad +2\sin^2\theta\cos^2\theta\left(1 - 2\sin^2\frac{(E_2 - E_1)t}{2}\right) && (103)\\
&= 1 - \sin^2 2\theta \sin^2\frac{(E_2 - E_1)t}{2} && (104)\\
&= 1 - \sin^2 2\theta \sin^2\frac{\Delta m^2_{21} t}{4E} && (105)
\end{aligned}
$$

Then we have disappearance probability:

$$P(\nu_\mu \to \nu_\mu, t) = |\langle \nu_\mu(t) \mid \nu_\mu(0) \rangle|^2 \quad (106)$$

$$= 1 - \sin^2 2\theta \sin^2 \left(\frac{\Delta m^2_{21}}{4E} t \right) \quad (107)$$

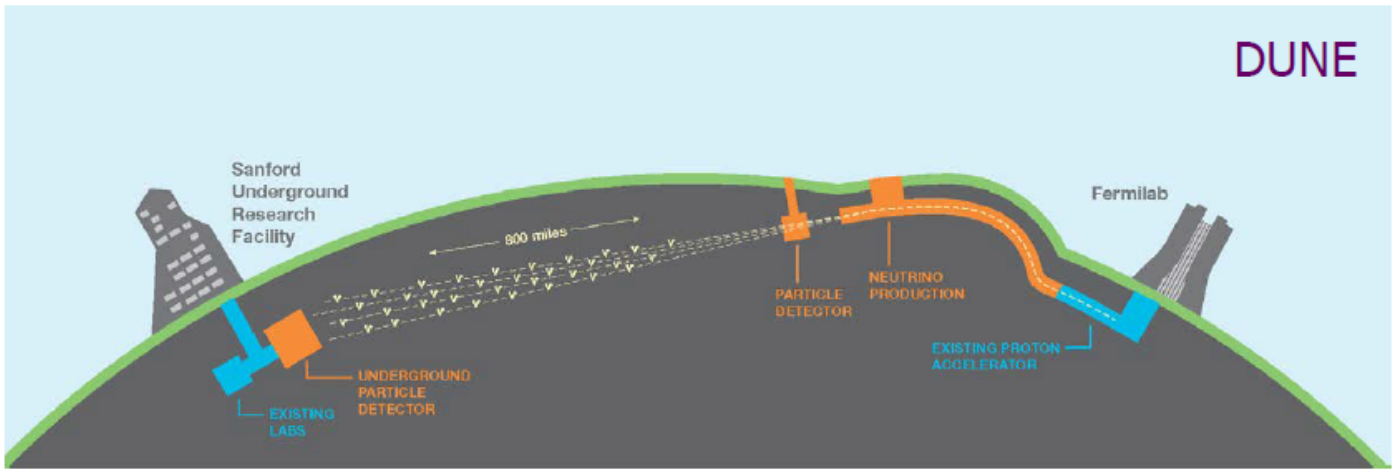
where $\Delta m^2_{21} = m^2_2 - m^2_1$ while

$$P(\nu_\mu \to \nu_e, t) = \sin^2 2\theta \sin^2 \left(\frac{\Delta m^2_{21}}{4E} t \right) \quad (108)$$

- let L denote the distance neutrinos travel in time t, so $L = t$ (for $c = 1$)
- The oscillation is maximal if

$$\left(\frac{\Delta m^2_{21}}{4E} L \right) = \frac{\pi}{2}$$

- Neutrino interactions are our doorway to neutrino properties:
  - Detection
    - Discovery: $\bar{\nu}_e\, p \rightarrow n\, e^+$ Cowan & Reines
  - Flavor ID: required to study oscillations
  - Neutrino propagation: matter effects (MSW)



  - Neutrino Astronomy